

Processing Speed and Reaction Time

Cognitive aging: APOE, KL-VS, BDNF, COMT, KIBRA, loci of reaction time and symbol-digital test (17q21/MAPT, PRDM11, RORB), fluid intelligence scores, mild cognitive impairment and cognitive fragility, and loci load

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. Sérgio F is **APOE ε3/ε3** (without ε4, the largest genetic factor of cognitive decline) and does not have the risk allele of **TOMM40**. The genetic fluid intelligence is at the upper limit of the typical (78.7 and 79.3 percentiles) and the genetic education (cognitive reserve) in the 65.7. In the reaction time loci there are 2 copies of the fastest reaction alleles in **KANSL1 (17q21/MAPT)** and **FOXO3**, and none of the slow allele of SEMA6D. The **KIBRA** has 1 allele of better memory.

Unfavourable. The score of mild cognitive impairment is in percentile 98.6, but the reliability is low (discords from the other scores; already seen in §34). The loci load of the reaction time is in percentile 78.8 (slowly to the slow side) and that of the general cognitive capacity in the 26.2. The cognitive fragility score is in the 76.6 (typical-high). It has BDNF Val66Met heterozygous (1 Met: a little less BDNF released with the activity), 2 alleles of ZNF184 (worst symbol-digit test), 2 locus APC and 1 in RORB and PRDM11.

What does it mean? The genetic basis of cognition is good and without the main accelerator of brain aging (ε4). The genetic reaction rate is typical, with signs in both directions. The processing speed depends mainly on what can be trained: aerobic capacity, sleep, vascular health, hearing and specific velocity training. The overall cognitive capacity and plasticity are in §34; the brain and neurodegeneration in §25.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Keyvariants** 15 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 8 of the chip, 7 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 1 SNPs needed conversion to the complementary strand and 1 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 5 PGS Catalog scores calculated locally with 3.510.720 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVÁVEL/ALTO; ≤ 20 : MENOS PROVÁVEL/BAIXO; between 20 and 80: TÍPICO.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 1495 loci, 1413 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In fluid intelligence and schooling, a high percentile = better. In cognitive fragility and mild cognitive impairment, a higher percentile = more risk.

Cognitive capacity

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Fluid intelligence	PGS002135 (Privé 2022, LDpred2, 903 thousand variants)	885.220 903.259 (weighted coverage 97%)	78,7	TIPIC	typical
Fluid intelligence	PGS001919 (Privé 2022, penalized, 26 thousand variants)	25.428 26.145 (weighted coverage 97%)	79,3	TIPIC	typical
University school (cognitive reserve)	PGS002685 (Weissbrod 2022, SbayesR, 981 thousand variants)	958.572 980.529 (weighted coverage 97%)	65,7	TIPIC	typical

Cognitive aging

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Cognitive fragility (5 factor of frailty)	PGS005226 (Foot 2025, SbayesR, 1.7 million variants)	1.605.399 1.715.891 (weighted coverage 89%)	76,6	TIPIC	typical
Mild cognitive deficit	PGS018931 (Chen 2025, Lassosum2, 38 thousand variants)	36.101 37.738 (weighted coverage 95%)	98,6	MAIS PROVABLE	unfavourable (low reliability)

- **Mild cognitive deficit:** Score trained in diagnosis of clinical records with few cases; disagrees with fluid intelligence scores and loci load (low reliability, as in §34)

4. Key variants by topic

Variants with effect on processing speed, reaction time and cognitive aging.

Cognitive reserve and brain aging

APOE is the gene with the greatest effect on cognitive aging: $\epsilon 4$ accelerates the decline of memory and processing speed from the 60 years. The klotho (KL-VS) and TOMM40 modulate this effect.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
APOE	rs429358	$\epsilon 4$ (Cys130Arg)	T/T	imputation DR2=0.99	C (0 copy/s)	favourable	Defines the $\epsilon 4$ allele (with rs7412): more memory decline and processing speed with age and more Alzheimer	Strong
APOE	rs7412	$\epsilon 2$ (Arg176Cys)	C/C	chip	T (0 copy/s)	neutral	Defines the $\epsilon 2$ (Arg176Cys) allele: slightly protective for cognitive decline	Strong
TOMM40	rs2075650		A/A	chip	G (0 copy/s)	favourable	APOE neighbor (mitochondrial import); it is associated with worse speed and memory, in part by binding to $\epsilon 4$	Moderate
KL (klotho)	rs9536314	KL-VS	T/T	chip	G (0 copy/s)	neutral	KL-VS (Phe352Val): in heterozygosity it is associated with more klotho, better cognition and less decline (Dubal 2014)	Moderate

Neurotransmission and plasticity

BDNF supports synaptic plasticity (and rises with exercise). COMT regulates prefrontal dopamine, which influences working memory and attention.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
BDNF	rs6265	Val66Met	C/T	chip	T (1 copy/s)	unfavourable (light)	Val66Met: Met (T) allele reduces activity-dependent BDNF secretion; episodic memory and plasticity slightly lower, especially with age	Moderate
COMT	rs4680	Val158Met	G/G	chip	A (0 copy/s)	without judgment	Val158Met: Met (A) = slower COMT and more prefrontal dopamine (working memory), Val (G) = faster; effect depends on context (stress, age)	Moderate
WWC1 (KIBRA)	rs17070145		C/T	chip	T (1 copy/s)	favourable	The T allele is associated with better episodic memory (Papestropoulos 2006)	Weak
CHRNA4	rs1044396		A/A	chip	T (2 copy/s)	without judgment	Nicotinic receptor $\alpha 4\beta 2$: visual attention and speed (small studies)	Weak

Loci of reaction time and velocity

The reaction time and the symbol-digital test measure the processing speed, the first cognitive domain to decline (from 30–40 years, ~ 0.02 standard deviations per year). Loci have very small effects.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
KANSL1 (17q21, MAPT)	rs7212857	C/C	imputation	DR2=0.99	C (2 copy/s)	favourable	17q21 block (MAPT/KANSL1), the strongest signal of reaction time: C allele is associated with faster reaction	Moderate
FOXO3	rs2153960	G/G	chip		G (2 copy/s)	favourable	FOXO3 (gene of longevity): the G allele is associated with faster reaction	Moderate
SEMA6D	rs16959783	T/T	imputation	DR2=0.94	C (0 copy/s)	favourable	Semaphorin 6D (axonal guide): C allele is associated with slower reaction	Weak
APC (LINC02200)	rs56335290	C/C	imputation	DR2=0.97	C (2 copy/s)	unfavourable	Locus near APC: C allele is associated with slower reaction	Weak
RORB	rs2604270	G/C	imputation	DR2=0.80	C (1 copy/s)	unfavourable (light)	Nuclear receptor RORB (sensory cortex): C allele is associated with slower reaction [GWAS Catalog: Reaction time (baseline), C allele, beta 0.020 [0.014-0.026] unit increase, p=2E-12, PMID 40374629]	Moderate
PRDM11	rs4486613	T/C	imputation	DR2=0.86	C (1 copy/s)	unfavourable (light)	Locus PRDM11: C allele is associated with the slower reaction [GWAS Catalog: Reaction time (baseline), C allele, beta 0.014 [0.0096-0.018] unit increase, p=1E-10, PMID 40374629]	Weak
ZNF184	rs35984974	G/G	imputation	DR2=1.00	G (2 copy/s)	unfavourable	Locus of the symbol-digit test: G allele is associated with worse performance (processing speed) [GWAS Catalog: Symbol digital substitution test performance (baseline), allele G, beta 0.038 [0.028-0.047] unit decrease, p=3E-14, PMID 40374629]	Weak

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of reaction time, symbol-digit test, general cognitive capacity and fluid intelligence, with the alleles oriented towards the highest value (in reaction time, higher = slower), compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Reaction time (slowest)	68 (62)	70 de 124	62	78,8	typical
Symbol-digit test (speed)	69 (60)	44 de 120	60	51,5	typical
General cognitive capacity	744 (713)	660 de 1426	710	26,2	typical
Fluid intelligence	614 (578)	530 de 1156	577	50,1	typical

6. Favourable

- **APOE ε3/ε3** (rs429358 T/T, rs7412 C/C): without ε4. **Strong.**
- **TOMM40 rs2075650 A/A**: without the APOE neighboring risk allele. **Moderate.**
- **Fluid intelligence in the upper limit of the typical**: percentiles 78.7 and 79.3; education (cognitive reserve) in 65.7. **Moderate.**
- **Faster reaction alleles**: KANSL1 rs7212857 C/C (block 17q21/MAPT, the strongest signal of reaction time) and FOXO3 rs2153960 G/G; SEMA6D without the slow allele. Moderate (small effects).
- **KIBRA rs17070145 C/T**: 1 allele with better episodic memory. **Weak.**
- **Symbol-digit test** (loci) in percentile 51.5 and fluid intelligence (loci) in the 50.1: typical. **Moderate.**

7. Unfavourable

- **Mild cognitive deficit:** score in percentile 98.6; low reliability (small cases in training; disagrees with fluid intelligence). **Weak.**
- **Reaction time** (loci) in percentile 78.8: slightly to the slower side, still typical. **Weak moderated.**
- **General cognitive capacity** (loci) in percentile 26.2; cognitive fragility in 76.6. **Weak.**
- **BDNF Val66Met C/T** (1 Met): less BDNF released with activity; small effect on memory and speed, higher with age and little exercise. **Moderate.**
- **ZNF184 rs35984974 G/G** (worst symbol-digit test), **APC rs56335290 C/C**, **RORB** and **PRDM11** (1 slow allele each). **Weak.**
- **KL-VS absent** (without the klotho bonus, which only ~25% of people have). **Neutral.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Proteostasis loss	APOE ε3/ε3 (without ε4) and block 17q21/MAPT (tau) with the fastest reaction alleles	Favourable
Protein Aggregation	Without ε4: less amyloid and tau with age	Favourable
Intercellular Communication	BDNF Val66Met (1 Met): less activity-dependent BDNF	Unfavourable (mild)
Nutrient Sensing Deregulation	FOXO3 rs2153960 G/G (faster reaction allele, longevity gene)	Favourable
Mitochondrial dysfunction	TOMM40 without risk allele (importation of mitochondrial proteins in neurons)	Favourable
Neuroinflammation	High but unreliable mild cognitive impairment score	Neutral
Vascular Aging	The processing speed depends on the white matter and the small vessels (see the stroke report and small vessels)	Neutral
Sleep disturbance	Deep sleep and glymphatic function support the processing speed; there are no specific risk genes here	Neutral

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Aerobic training (zone 2 + intervals):** aerobic capacity is the best modifiable predictor of processing speed and increases BDNF, which partially compensates Val66Met. *Suggestion (not applied).*
2. **Strength and balance/agility training** (Racket sports, coordination, direction changes): the reaction time of choice predicts falls and fragility. *Suggestion (not applied).*
3. **Specific processing speed training** (useful visual field exercises, such as ACTIVE, 10–20 sessions with reinforcements): improved speed during 10 years and was associated with less dementia. *Suggestion (not applied).*
4. **Measure the baseline itself:** simple reaction time and a symbol-digit test, 1–2 times per year, under the same conditions (morning, rested). A sustained descent is a useful early signal. *Suggestion (not applied).*
5. **7–9 h sleep with regular time:** a short night worsens the reaction time as well as several years of aging. *Suggestion (not applied).*
6. **Hearing and vision:** to correct early hearing loss (see the Audition and Zombido report) and vision; sensory loss is one of the major modifiable factors of cognitive decline. *Suggestion (not applied).*
7. **Vascular brain health:** LDL and ApoB as low as possible, stable glucose, zero tobacco and zero alcohol (already fulfills); Omega-3 3.2–4 g/day of the plane (DHA for neuronal membranes). *Suggestion (not applied).*
8. **Caffeine in moderate dose before midday** improves the reaction time in an acute way; creatine (if already in the plane) helps speed especially in sleep deprivation. *Suggestion (not applied).*

10. Limitations

- The reaction time loci explain a very small fraction of the variation; training, sleep and age weigh much more.
- The mild cognitive impairment score was trained with few cases of clinical records and disagrees with the others; it should not be read alone.
- The 17q21 block (MAPT/KANSL1) is an inversion with H1/H2 haplotypes; the isolated SNPs are only markers.
- The chip does not read rare variants of early dementia (PSEN1, PSEN2, APP, GRN, C9orf72).
- The load of SNPs in linkage disequilibrium is descriptive.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the Imposed VCF; in the palindromics the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Polygenic scores:** Harmonized GRCh38 files from the PGS Catalog (5 scores: fluid intelligence, education, cognitive fragility and mild cognitive impairment). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 510^{-8}$), alleles guided by the published effect signal, empirical calibration in the same reference population.
- **Interpretation:** reference literature (Davies 2018 for reaction time; Dubal 2014 for KL-VS; Egan 2003 for BDNF; Papassotyropoulos 2006 for KIBRA; Rebok 2014 and Edwards 2017 for the ACTIVE trial, GWAS Catalog and ClinVar. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Cognitive reserve and brain aging	APOE	rs429358	T/T	imputation DR2=0.99	C	0	strand +	without ε4
Cognitive reserve and brain aging	APOE	rs7412	C/C	chip	T	0	strand +	without the protective allele (common genotype)
Cognitive reserve and brain aging	TOMM40	rs2075650	A/A	chip	G	0	strand +	without the risk allele
Cognitive reserve and brain aging	KL (klotho)	rs9536314	T/T	chip	G	0	strand +	without KL-VS (common genome)
Neurotransmission and plasticity	BDNF	rs6265	C/T	chip	T	1	strand +	1 copy of risk allele
Neurotransmission and plasticity	COMT	rs4680	G/G	chip	A	0	strand +	0 copy of the effect allele
Neurotransmission and plasticity	WWC1 (KIBRA)	rs17070145	C/T	chip	T	1	strand +	1 copy of the protective allele
Neurotransmission and plasticity	CHRNA4	rs1044396	A/A	chip	T	2	complement (strand)	2 copy of the effect allele
Loci of reaction time and velocity	KANSL1 (17q21, MAPT)	rs7212857	C/C	imputation DR2=0.99	C	2	strand +	2 copies of the protective allele
Loci of reaction time and velocity	FOXO3	rs2153960	G/G	chip	G	2	strand +	2 copies of the protective allele
Loci of reaction time and velocity	SEMA6D	rs16959783	T/T	imputation DR2=0.94	C	0	strand +	without the risk allele
Loci of reaction time and velocity	APC (LINC02200)	rs56335290	C/C	imputation DR2=0.97	C	2	strand +	2 risk allele copies
Loci of reaction time and velocity	RORB	rs2604270	G/C	imputation DR2=0.80	C	1	strand + (palindromic; assumed +)	1 copy of risk allele
Loci of reaction time and velocity	PRDM11	rs4486613	T/C	imputation DR2=0.86	C	1	strand +	1 copy of risk allele
Loci of reaction time and velocity	ZNF184	rs35984974	G/G	imputation DR2=1.00	G	2	strand +	2 risk allele copies